

Priyadarshini College of Engineering, Nagpur
Department of Computer Technology
Session 2023-24
Question Bank
Subject: Blockchain and it's Applications

Unit 1

- What is Decentralization? Explain with the help of supply chain management example.
- Explain the importance of hashing with respect to blockchain
- Explain how and why SHA 256 is used in Blockchain
- Discuss SHA256 Algorithm with processing.
- What is Hash Pointer? Discuss temper detection using hash chain
- Explain Markle tree and structure of Markle tree
- Discuss RSA algorithm for key generation and verification with an example.
- Alice employs the RSA cryptosystem with the prime numbers $P=7$ and $Q=17$ to derive her public and private keys. Given that Alice's public key is 11 , what will be her private key?
- What is Digital Signature? Explain
- Explain Cryptocurrencies and its evolution.
- Wxplain Evolution of blockchain

Unit 2

- Discuss DApps.
- What are smart Contracts.Explain various types of smart contracts
- What is Blockchain.Draw and Explain structure of Block and block header.
- What are Bitcoin scripts? Explain with example.
- What are the steps for joining a bitcoin network? Explain
- What is double Spending? Explain how double spending is handled using blockchain.

Unit 3

- What is block mining? desribe the role of miners in detail.
- Explain Bitcoins
- Differentiate Permissioned and Permission less Blockchain
- Discuss
 - I. Proof of Work(PoW) II. Proof of Stack III. Proof of Burn
 - IV. Proof of Elapsed Time
- What is Nonce? How is it used to solve puzzles in blockchain?
- Differentiate Proof of Work and Proof of Stake
- Discuss Smart Contracts over Closed Networks
- What are the various Faults in a Distributed Systems? Explain
- Explain Byzantine Faults and Byzantine Agreement Protocols

Unit 4

- Draw and explain Ehthereum Network.

- Discuss DAPPS in Ethereum
- Discuss Ethereum Virtual Machine - EVM
- Discuss Ethereum Smart Contracts
- Solidity
- Explain and list the part of EVM Memory in blockchain.
- What is Hyperledger fabric? Explain in detail
- Explain Hyperledger sawtooth.
- Draw and explain fabric architecture.
- Differentiate between PoW Consensus vs BFT Consensus
- Explain the blockchain performance triangle.
- What are the function modifiers in solidity? How is blockchain distributed ledger different from the traditional ledger?

Unit 5

- What is Identity? Explain Centralized Digital Identity
- Explain Decentralizing Digital Identity
- Discuss Decentralized Identifiers (DIDs)
- Explain in detail how can blockchain be used for identity management.
- Discuss about the emerging trends in blockchain technology and its applications in real time scenarios.
- Discuss problem with interoperability in blockchain using an example
- Explain how blockchain can be used in financial services